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We lossen op in $\mathbb{C}$: $2z(z+1) + z^2 + 1 = 0$

$$\Leftrightarrow 2z^2 + 2z + z^2 + 1 = 0$$

$$\Leftrightarrow 3z^2 + 2z + 1 = 0$$

$$D = b^2 - 4ac = 2^2 - 4 \cdot 3 \cdot 1 = 4 - 12 = -8$$

$$\sqrt{D} = \sqrt{-8} = \sqrt{8}i = 2\sqrt{2}i$$

$$z = \frac{-2 \pm 2\sqrt{2}i}{6} = \frac{-1 \pm \sqrt{2}i}{3}$$

$$\text{Oplossingen: } z_1 = \frac{-1 + \sqrt{2}i}{3} \text{ en } z_2 = \frac{-1 - \sqrt{2}i}{3}$$

References